Unreliability or Disagreement? What an Interpercentile Decomposition Can and Cannot Show

Anonymous Author(s)
Affiliation
Address
email

Abstract

A widely cited result reports that language models "get lost" in multi-turn conversation, and decomposes the degradation into a small loss of *aptitude* and a large increase in *unreliability*, where aptitude is a high percentile of an instruction's score distribution and unreliability is an interpercentile range. We show that this decomposition cannot support the interpretation usually placed on it. The underlying statistics are not new. Dispersion of a binary variable peaks at intermediate difficulty, a fact long known in classifier-ensemble diversity and item-response theory. They have not, however, been applied to this metric, and their consequences here are severe. Analytically: the two components are nested, since the range is the difference of the two percentiles that define aptitude and the lower tail, so they are one reparameterisation of a percentile pair rather than independent axes; at the original's own ten runs per instruction and under a binary scorer, the range saturates at its maximum for seven of eleven possible outcomes and vanishes at both extremes, making it a near-binary indicator of whether an instruction is contested rather than a graded measure of reliability, so a model that always fails is scored as reliably as one that always succeeds; and because a range is non-negative, the comparison degenerates as a baseline saturates: the reported change is forced non-negative in the limit, and its relative form is inflated well before that. Empirically, on 5,765 conversations we generated across seven models from six developers and three tasks, three of sixteen cells have a baseline unreliability of exactly zero, so their headline increase is guaranteed by construction; a null containing no reliability parameter at all overproduces the original's headline unreliability rise while underproducing its aptitude drop, inverting the intended reading; and a binomial null that sees only each cell's accuracies and run count, and nothing about the task, recovers the ordering of the per-task results (Spearman $\rho = 0.66$). Replacing the statistic with an operational retry test separates what the decomposition conflates: on mathematics an oracle retry recovers 92% of the degradation, while on function calling it recovers 48% and leaves a gap strictly positive in seven of seven cells, with a fifth of instructions failing deterministically. We conclude that interpercentile decompositions should be reported alongside the percentiles they are built from, and that benchmark saturation must be checked before such a decomposition is interpreted.

1 Introduction

Empirical claims about language models are usually claims about a summary statistic. When that statistic has structure the reader does not see, a conclusion can be an artefact of the construction rather than a fact about models. This paper is a case study.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

We take as our object of study the analysis of Laban et al. [9], which reports that models degrade sharply when an instruction is revealed over several turns rather than stated at once, and decomposes that degradation into *aptitude* A_h, the hth percentile of an instruction's score distribution, and *unreliability* $U_l^h = A_h - P_l$, the gap between the hth and lth percentiles. The reported conclusion is that aptitude falls only slightly while unreliability rises sharply, which invites the reading that models remain capable but become noisy.

The degradation itself is robust and we reproduce it in every condition we ran. Our contribution concerns the *decomposition*. We show that it cannot distinguish the interpretation it is given from several alternatives, for reasons that are partly analytic and partly empirical, and we quantify how much of an apparent per-task pattern survives a null model that knows nothing about tasks.

Contributions. (i) Three structural properties of the interpercentile decomposition: nesting, non-monotonicity under coarse scorers, and degeneracy as a baseline saturates. (ii) A reproduction on 5,765 self-generated conversations across seven models from six developers and three tasks, showing all three properties bind in practice. (iii) A binomial null that recovers the ordering of the per-task results without any task information, indicating that per-task differences in this decomposition should not be read as facts about tasks.

Related work. That the dispersion of a binary variable is maximal at intermediate difficulty and vanishes at both extremes is long established. Kuncheva and Whitaker [8] define the Kohavi–Wolpert variance for classifier ensembles with exactly this shape and tabulate invariance under exchanging success and failure as a property of diversity measures; Gulliksen [7] makes the corresponding point for test items, and item-response theory's information function has the same form. The standard remedy is to correct observed agreement against the agreement expected by chance at the observed base rate, as Fleiss's κ does [6], and the associated base-rate paradoxes are well documented [4]. Closer to our setting, Bowyer et al. [2] show that normal-approximation intervals collapse spuriously as accuracy approaches zero or one in LLM evaluation, and Clark et al. [3] note that a model can be trivially self-consistent by answering uniformly. Our contribution is not the underlying statistics, which are textbook, but their application to a specific and widely cited evaluation metric: what those facts imply for an interpercentile decomposition, how severe the degeneracy is at the run counts actually used, and an empirical demonstration on a reproduction. Evaluations elsewhere face the same pressure to fix an arbitrary analysis parameter: emergent-misalignment work gates generations on a coherence score with a threshold its authors call arbitrary, and then checks that the pattern of results survives varying it [1]; downstream work adopts the same thresholds [10]. That is the practice we argue for, and it is what the decomposition we study lacks.

2 Three properties of the decomposition

Let an instruction be run N times, giving scores $S = \{S_i\}_{i=1}^{N}$. Following Laban et al. [9],

$$P \;=\; \mathrm{mean}(S), \qquad A_h \;=\; \mathrm{percentile}_h(S), \qquad U_l^h \;=\; \mathrm{percentile}_h(S) - \mathrm{percentile}_l(S), \tag{1}$$

with deltas throughout being multi-turn minus single-turn. The original writes these as A^{90} and U_{10}^{90}; we keep h and l as free symbols so that the percentile convention can be varied.

P1: the components are nested. By definition $U_l^h = A_h - P_l$, so $\Delta U = \Delta A - \Delta P_l$. Writing the contrast that the qualitative claim relies on as $m = \Delta U - |\Delta A|$ and substituting, for any cell with $\Delta A \le 0$,

$$m \;=\; (\Delta A - \Delta P_l) + \Delta A \;=\; 2\,\Delta A - \Delta P_l. \tag{2}$$

The comparison of "unreliability against aptitude" is therefore a comparison of the lower tail against twice the upper tail. The sharper statement is sufficiency: for N exchangeable binary trials the success count is a sufficient statistic [5], so A_h, P_l, U_l^h, the mean and the variance are all deterministic functions of a single number, and no exchangeable summary of that sample can separate capability from consistency. It is one reparameterisation of the percentile pair (P_h, P_l), not two independent measurements, and a confidence interval on ΔA and one on ΔU are not independent either. We verified Eq. 2 to numerical precision in all 15 of our cells with $\Delta A \le 0$.

P2: under a coarse scorer, unreliability measures disagreement, not reliability. When the scorer is binary, S is determined by the success count k, and so is U_l^h. The original analysis uses $N = 10$ simulations per instruction, where U_{10}^{90} takes the values $0, 0.1, 1, 1, 1, 1, 1, 1, 0.1, 0$ for $k = 0, \ldots, 10$: it is very nearly an *indicator* of whether between two and eight of the ten runs succeeded. At our $N = 6$ it is $0, \frac{1}{2}, 1, 1, 1, \frac{1}{2}, 0$. So it saturates at its maximum across seven of the eleven possible outcomes, and $\mathbb{E}[U \mid p]$ is confined to $[0.863, 0.980]$ across the whole middle of the ability range. When $h + l = 1$ the statistic obeys $U(k) = U(N - k)$ exactly, so $\mathbb{E}[U \mid p]$ is symmetric about $p = 1/2$ and attains its maximum there: $\mathbb{E}[U \mid 0.3]$ and $\mathbb{E}[U \mid 0.7]$ are the same number, 0.8627. It is therefore not merely low-resolution but non-monotone, and cannot even sign the direction of an ability change: a rise in U is equally consistent with a model improving toward $p = 1/2$ or degrading toward it. In both cases it is *non-monotone* and vanishes at $k = 0$ and $k = N$: a model that fails every run is recorded as exactly as reliable as one that succeeds every run. Under a binary scorer the reported doubling of "unreliability" therefore says that the share of *contested* instructions roughly doubled, which is a weaker and different statement. Figure 1(b) shows $\mathbb{E}[U \mid p]$ under $\mathrm{Binomial}(N, p)$, an inverted U peaking at $p = \frac{1}{2}$. The statistic answers "is this instruction contested?", which coincides with reliability only away from the extremes. Consequently ΔU is largely determined by how far accuracy moves along that curve, and a model that degrades from perfect to merely inconsistent and one that degrades from inconsistent to uniformly failing produce changes of opposite sign for the same drop in accuracy. These values use linear interpolation between order statistics (the NUMPY default, Hyndman–Fan type 7); under an inverse-ECDF convention the saturation is wider still, covering eight of eleven outcomes. The obvious repair, rescaling the statistic by what is attainable at the observed base rate in the spirit of chance-corrected agreement coefficients [6], does not work here. For binary scores the plug-in ratio is $s^2/(\hat{p}(1-\hat{p})) = N/(N-1)$ *identically* for every $0 < k < N$, because the normaliser is a bijective function of the numerator. At $N = 10$ it equals 1.111 for every outcome: base-rate normalisation does not restore resolution, it removes what little remained.

P3: as a baseline saturates, the comparison degenerates. U_l^h is a range, so $U_l^h \geq 0$. In the limit where the baseline condition is saturated, $U^{\text{single}} = 0$ and therefore $\Delta U = U^{\text{multi}} \geq 0$ whatever the model does: the metric cannot report a reliability improvement even in principle, and the *relative* change diverges. The original's baselines are not at that limit: its single-turn unreliability is well away from zero, so its reported sign is not mechanically forced. Its relative form is nonetheless denominator-sensitive: the four of its models with the smallest single-turn unreliability show relative rises of 236% to 285%, against a corpus-level $+112\%$ computed on means. It is our reproduction, not the original, that reaches the degenerate regime, with single-turn unreliability from 0.0 to 20.4 points against the original's 13 to 49. The authors note a related effect for weak models, that lower Full scores leave "less scope for degradation" [9], but treat it as a caveat about model comparison rather than about the decomposition, and their limitations section raises no metric concern. Our reproduction reaches the degenerate limit itself: three of sixteen cells have U^{single} exactly zero.

3 Setup

The authors release sharded instructions but not their conversation logs, so all conversations here were generated. Four of the original's six tasks are scored for binary correctness before being mapped onto a 0–100 range [9], so the coarse-scorer regime of P2 is the one its headline numbers largely occupy. We use three tasks with programmatic scorers, so that no model-graded judge enters the pipeline: *mathematics* (exact numeric match, 30 instructions), *function calling* (abstract-syntax-tree match, 25 instructions) and *data-to-text* (BLEU against multiple references, 25 instructions). Conversations were generated through a serverless inference API for Llama 4 Maverick, Llama 3.3 70B, Qwen3-32B, DeepSeek R1, Claude Sonnet 4.5, Mistral Pixtral Large and Amazon Nova 2 Lite. The frozen corpus holds 7,048 lines, of which 7,047 parse and 6,794 carry a score (one line is truncated and is excluded); 5,765 conversations fall in the 16 model-task cells that pass our power gates.

Because A_h and U_l^h are percentiles taken *within* an instruction, runs per instruction is the binding constraint, not instruction count. We analyse a cell only if it has at least 10 instructions paired across both conditions and a median of at least 4 runs per instruction per condition; 5 of 21 cells fail these

gates and are excluded and listed, not silently dropped. Both conditions are always evaluated on an identical instruction set. Confidence intervals are bootstrap intervals over instructions ($B = 2000$).

4 Results

The degradation reproduces. Multi-turn performance falls in every cell, by -5.9 to -65.3 percentage points. We do not dispute the original headline finding.

Aptitude is a threshold readout, not a capability measure. The same reduction applies to A_h. At $N = 10$, $\mathbb{E}[A_{90} \mid p]$ crosses $1/2$ at $p = 0.153$ and exceeds 0.998 for every $p \geq 0.6$: it is a smoothed indicator of $p > 0.15$, with no resolution across the range where capable models operate. Aptitude and unreliability are thus two threshold readouts of one instruction-difficulty distribution, taken at different cut points, rather than two axes of model behaviour.

A null with no reliability parameter overproduces the headline effect. We can ask directly how much of the reported pattern survives removing reliability from the data-generating process. Draw per-instruction success probabilities $p_i \sim \mathrm{Beta}$ calibrated to a single-turn accuracy near ceiling and a multi-turn drop of comparable size to the original's reported degradation, and let the N runs of an instruction be independent $\mathrm{Bernoulli}(p_i)$. This model has no reliability parameter at all: every run is exchangeable and independent given the instruction. Across concentrations from 2 to 30 it yields ΔU between $+167\%$ and $+196\%$, against the reported $+112\%$, and ΔA between -0.2% and -6.9%, against the reported -16%. The reported rise in unreliability is therefore *smaller* than a model with no reliability structure produces, while the reported fall in aptitude is *larger*. Read against the right null, the anomalous component of the degradation is the loss of capability, not the growth of variance, which is the opposite of the conclusion the decomposition is used to support.

Read correctly, the original's own statistic shows capability collapse. Because $\mathbb{E}[A_h \mid p]$ is increasing, for any difficulty distribution G and any t we have the distribution-free bound $G([0, t]) \geq \Phi_A(t) - \bar{A}$, where Φ_A is the transfer function above. The original reports a 16% relative fall in aptitude, so its multi-turn $\bar{A} \leq 0.84$. The bound then forces at least 15% of instructions below a multi-turn success probability of 0.5, and at least 16% below 0.7. That is a hard capability loss on a sixth of the instruction set, derived from the very statistic offered as evidence that capability was largely preserved.

The degenerate limit is reached in practice. Table 1 reports each cell's baseline unreliability U^{single} alongside the decomposition. Seven of sixteen cells have $U^{\text{single}} < 5$pp, and in all seven $\Delta U > 0$. In three cells U^{single} is *exactly* zero, so their reported increases of $+20.0$, $+30.0$ and $+44.4$ points are non-negative by construction and carry no evidence about reliability.

The original's relative claim is more robust than its absolute form. The original reports *relative* changes (aptitude -16%, unreliability $+112\%$). Because U^{single} is small whenever the baseline is near ceiling, a small absolute rise in a small denominator yields a large percentage. Evaluated in relative form, the ordering holds in 14 of our 16 cells, against 10 of 16 in absolute form. The relative statement is thus better supported, but it is also the form most inflated by a saturated baseline, so the two parameterisations should be reported together.

A task-blind null recovers the per-task ordering. If one groups cells by task, the contrast m has median $+23.3$ on mathematics, $+0.0$ on function calling and -8.5 on data-to-text, which invites a claim that the decomposition is task-dependent. We test that against a null in which each cell is a homogeneous binomial with only that cell's single-turn and multi-turn accuracy and its run count, and no information about the task. The null reproduces the ordering of the observed margins (Spearman $\rho = 0.66$, Figure 1(c)), while badly misestimating their magnitudes, as expected since real instructions are heterogeneous. The qualitative per-task pattern is therefore substantially recoverable from accuracy and run count alone, and should not be read as a fact about tasks.

An operational test: is the loss actually recoverable? The decomposition's practical promise is falsifiable. If capability is intact and only consistency is lost, then *retrying* should recover the loss.

Table 1: Decomposition at the $90/10$ convention on our frozen corpus. U^{s} is single-turn unreliability. Cells with $U^{\mathrm{s}} = 0$ (marked †) have $\Delta U \geq 0$ by construction. $m = \Delta U - |\Delta A|$.

Task	Model	U^{s}	ΔP	ΔA	ΔU	m
Mathematics	Llama 4 Maverick	10.0	−35.0	−10.3	+37.7	+27.3
Mathematics	Nova 2 Lite	6.7	−37.1	−15.0	+40.7	+25.7
Mathematics	Qwen3-32B	16.3	−33.6	−18.0	+41.3	+23.3
Mathematics	DeepSeek R1	6.7	−28.6	−10.3	+33.3	+23.0
Mathematics	Claude Sonnet 4.5	1.7	−26.2	−13.3	+26.7	+13.3
Function calling	Nova 2 Lite	0.0†	−53.1	−32.0	+44.4	+12.4
Function calling	Qwen3-32B	2.0	−46.8	−30.0	+36.4	+6.4
Function calling	Llama 4 Maverick	0.0†	−39.3	−26.0	+30.0	+4.0
Function calling	Llama 3.3 70B	0.0†	−30.7	−20.0	+20.0	+0.0
Function calling	DeepSeek R1	2.0	−65.3	−45.6	+38.8	−6.8
Function calling	Claude Sonnet 4.5	5.3	−41.0	−29.5	+22.1	−7.4
Function calling	Mistral Pixtral L	20.0	−32.8	−27.6	+9.5	−18.1
Data-to-text	Llama 4 Maverick	9.4	−5.9	+0.1	+11.7	+11.6
Data-to-text	Llama 3.3 70B	5.3	−13.9	−10.6	+5.8	−4.8
Data-to-text	DeepSeek R1	14.4	−10.2	−10.9	−1.4	−12.3
Data-to-text	Qwen3-32B	17.3	−9.4	−11.0	−3.3	−14.4

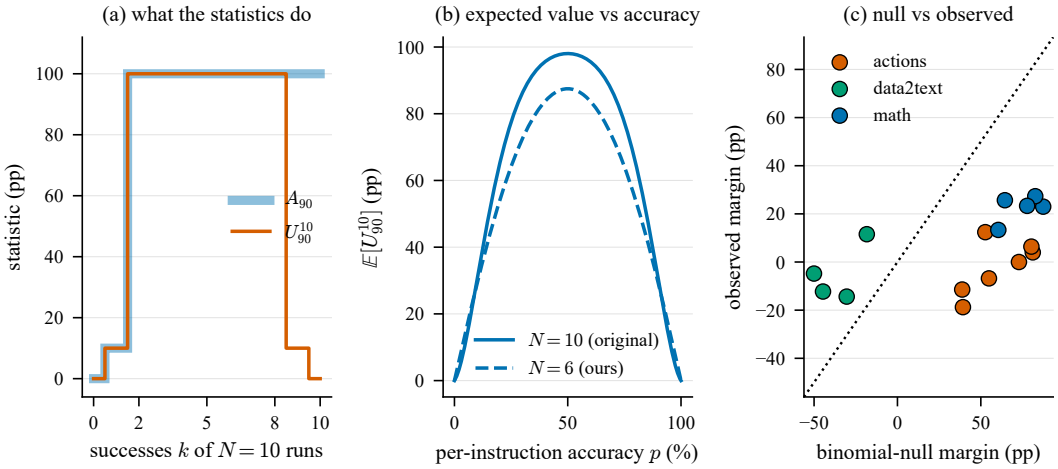


Figure 1: (a) With a binary scorer at the original's $N = 10$, both statistics are step functions of the success count k: A_{90} is a threshold indicator, and U_{10}^{90} saturates at its maximum for $k = 2 \ldots 8$ and vanishes at both ends, so a model that always fails scores as reliably as one that always succeeds. (b) Expected unreliability against per-instruction accuracy at $N = 10$ (solid) and our $N = 6$ (dashed); it is non-monotone and symmetric about $p = 1/2$. (c) Observed per-cell contrast m against a binomial null using only each cell's accuracies and run count, with no task information; the dotted line is equality. The null recovers the ordering ($\rho = 0.66$) while misestimating magnitudes.

We test this directly, counting an instruction as solved under a condition if at least one of its runs succeeds. This is an oracle: it assumes a perfect verifier that can pick the successful attempt, so it upper-bounds anything a deployment could achieve.

On mathematics the promise holds (Table 2). Oracle retry recovers a median 91.8% of the single-to-multi gap across five models, and the residual unrecoverable gap of 10.0 points has a bootstrap interval covering zero in five of five cells. Multi-turn mathematics really is a consistency problem.

On function calling it fails. Oracle retry recovers only 47.8% of the gap, and the unrecoverable remainder is 23.8 points with a bootstrap interval strictly above zero in *seven of seven* cells. A median 20.0% of instructions are *deterministic* failures: solved in every single-turn run and in none of the multi-turn runs. No number of retries recovers them, because there is nothing inconsistent about them: the capability is simply absent once the instruction is sharded.

Table 2: Operational retry test. Gap is single-turn minus multi-turn mean score. Recovered is the share of that gap closed by an *oracle* best-of-K over the multi-turn runs, which upper-bounds any deployable strategy. Unrecoverable is the residual gap in instructions solved at least once single-turn but never multi-turn, with a bootstrap interval over instructions; * marks intervals strictly above zero. Deterministic failures are solved in *every* single-turn run and in *no* multi-turn run.

Task	Model	Gap	Recov. (%)	Unrec. (pp)	95% CI	Determ. (%)
Mathematics	Llama 4 Maverick	35.0	108.2	3.3	$[0.0, 10.0]$	0.0
Mathematics	DeepSeek R1	28.6	98.5	3.3	$[0.0, 10.0]$	3.3
Mathematics	Qwen3-32B	33.6	91.8	10.0	$[0.0, 23.3]$	3.3
Mathematics	Nova 2 Lite	37.1	80.5	10.0	$[-3.3, 23.3]$	10.0
Mathematics	Claude Sonnet 4.5	26.2	64.0	10.0	$[-3.3, 23.3]$	13.3
Function calling	Qwen3-32B	46.8	58.7	20.0^*	$[7.9, 36.0]$	20.0
Function calling	Mistral Pixtral L	32.8	53.8	23.8^*	$[9.5, 42.9]$	14.3
Function calling	Llama 4 Maverick	39.3	49.2	20.0^*	$[4.0, 36.0]$	20.0
Function calling	Llama 3.3 70B	30.7	47.8	16.0^*	$[4.0, 32.0]$	16.0
Function calling	Nova 2 Lite	53.1	47.2	28.0^*	$[8.0, 48.0]$	32.0
Function calling	DeepSeek R1	65.3	45.9	36.0^*	$[16.0, 56.0]$	36.0
Function calling	Claude Sonnet 4.5	41.0	40.0	26.3^*	$[5.3, 47.4]$	26.3

This is the distinction the decomposition is used to make, and it is the distinction it cannot make. Both task families show a large drop in P and a large rise in U; only the retry test separates the case where capability survives from the case where it does not. It also costs nothing extra: it reuses the same runs, and unlike U_l^h it is monotone in capability, has an unambiguous estimand, and states a quantity a practitioner can act on.

5 Discussion

None of this shows that models do not degrade in multi-turn conversation. They do, in every cell we ran, and that finding is not in question. What the decomposition cannot do is license the reading that capability is preserved while reliability falls. By P1 the two quantities are one reparameterisation of a percentile pair; by P2 the reliability term is a disagreement rate that falls to zero when a model fails consistently; and by P3 its reported direction is fixed in advance whenever the baseline is saturated, which our data show is the common case.

The practical stake is not rhetorical. A practitioner told that models are "as capable but less reliable" will reach for sampling or self-consistency. In our function-calling cells, aptitude falls by 20 to 46 points, so best-case capability is genuinely lost and repeated sampling addresses the smaller part of the problem.

Recommendations. Report P_h and P_l directly alongside any interpercentile range; report the baseline value of the range, since a reader cannot otherwise tell a forced sign from an informative one; state the percentile convention and the number of runs per item, both of which set the resolution of the estimate; prefer a scorer with more than two levels when the object of study is variability; and do not expect base-rate normalisation to repair a within-instruction dispersion statistic, since for binary scores it is constant. A consistency claim needs either graded scores with at least three levels, a second replication factor against which extra-binomial variation can be tested, or an order-dependent statistic.

Limitations. Our tasks use three different scorer types, so per-task magnitudes are not comparable across tasks; this is one reason we argue the per-task reading is unsafe rather than proposing a corrected per-task claim. Our data-to-text condition does not invoke the LLM user-simulator, so it carries no simulator variance, unlike the other two tasks. Most cells use six runs per instruction, which places A_{90} and U_{10}^{90} on a three-point lattice under a binary scorer and limits per-cell resolution to $50/M$ points for M instructions; margins should be read at that resolution. We study open-weight and API models different from the original, so absolute levels are not comparable to it. The binomial null is a homogeneous approximation and is offered as a check on the per-task reading, not as a model of the data.

Data and reproducibility. The frozen corpus of 7,048 records, the analysis scripts and a README with the exact regeneration commands are released as supplementary material. All numbers here come from that frozen snapshot.

References

[1] J. Betley, D. Tan, N. Warncke, A. Sztyber-Betley, X. Bao, M. Soto, N. Labenz, and O. Evans. Emergent misalignment: narrow finetuning can produce broadly misaligned LLMs. *ICML*, 2025. arXiv:2502.17424. Extended version in *Nature*, 2026, doi:10.1038/s41586-025-09937-5.

[2] S. Bowyer, L. Aitchison, and D. R. Ivanova. Position: Don't use the CLT in LLM evals with fewer than a few hundred datapoints. *ICML*, 2025. arXiv:2503.01747.

[3] P. Clark, B. Dalvi Mishra, and O. Tafjord. BaRDa: a belief and reasoning dataset that separates factual accuracy and reasoning ability. *arXiv:2312.07527*, 2023.

[4] A. R. Feinstein and D. V. Cicchetti. High agreement but low kappa: I. the problems of two paradoxes. *Journal of Clinical Epidemiology*, 43(6):543–549, 1990.

[5] R. A. Fisher. On the mathematical foundations of theoretical statistics. *Philosophical Transactions of the Royal Society A*, 222:309–368, 1922.

[6] J. L. Fleiss. Measuring nominal scale agreement among many raters. *Psychological Bulletin*, 76(5):378–382, 1971.

[7] H. Gulliksen. The relation of item difficulty and inter-item correlation to test variance and reliability. *Psychometrika*, 10(2):79–91, 1945.

[8] L. I. Kuncheva and C. J. Whitaker. Measures of diversity in classifier ensembles and their relationship with the ensemble accuracy. *Machine Learning*, 51:181–207, 2003.

[9] P. Laban, H. Hayashi, Y. Zhou, and J. Neville. LLMs get lost in multi-turn conversation. *arXiv:2505.06120*, 2025.

[10] M. Taylor, J. Chua, J. Betley, J. Treutlein, and O. Evans. School of reward hacks: hacking harmless tasks generalizes to misaligned behavior in LLMs. *arXiv:2508.17511*, 2025.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction claim a reproduction of the degradation and task-dependence of its decomposition, which is exactly what the results report, including the cells where the original characterisation holds.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: A dedicated Limitations paragraph discusses scorer heterogeneity across tasks, the mix of open-weight and API models, the absence of simulator variance in one task, and the coarse percentile resolution at six runs per instruction.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper contains no theoretical results; the monotonicity observation about coupled percentile quantities is arithmetic, not a theorem.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: The Setup section specifies tasks, scorers, instruction counts, runs per condition, models, decomposition definitions and conventions; the simulation harness is the original authors' public code.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).

(d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: All scored conversation logs and analysis scripts are released as supplementary material, as stated in the data and reproducibility paragraph.

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
   - At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
   - Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: No models are trained. All inference settings (sampling counts, conversation protocol, scorer definitions) are given in the Setup section.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: The retry table reports bootstrap confidence intervals resampled over instructions (B=2000), the unit at which conditions are paired. Table 1 reports point estimates only; its intervals are included in the released results file.

   Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: All inference ran through a commercial serverless inference API; no local GPUs were used. Total volume is the conversation counts reported per cell.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: The work re-analyses published benchmark tasks with API-served models and conforms to the Code of Ethics.

   Guidelines:

   - The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
   - The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [N/A]

Justification: This is a methodological reproduction about evaluation reporting; it has no direct deployment pathway beyond encouraging convention disclosure.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: No models or scraped datasets are released; the released artefacts are scored conversation logs on public benchmark instructions.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: The original study, its released code and the underlying benchmarks are cited, and the harness is used under its published license.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.

- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: The released logs and scripts are documented in an accompanying README describing formats and regeneration commands.

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: The paper involves no crowdsourcing or human subjects; the simulated user is a language model.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
    - According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

Justification: No human participants were involved, so no IRB approval was required.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: LLMs are the object of study, and a fixed LLM user-simulator is part of the experimental method, described in the Setup section.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.